Unit 8 Guided Notes — Transformations

Mr. Allen's 8th Grade Math | NC.8.G.2 · NC.8.G.3 · NC.8.G.4

Name: _____ **Date:** _____ **Period:** _____

A **transformation** is a rule that moves every point of a figure. In this unit, you will learn four kinds: **translations** (slides), **reflections** (flips), **rotations** (turns), and **dilations** (resizes). You will also decide if two figures are **congruent** (same size and shape) or **similar** (same shape, different size).

Every transformation in this unit happens on the coordinate plane. Keep your pencil and grid ready.

1. Transformation Vocabulary

Fill in each blank as we work through the definitions.

Transformation: A rule that _____ every point of a figure in a predictable way.

Pre-image: The figure _____ the transformation. Vertices use plain letters: A, B, C.

Image: The figure _____ the transformation. Vertices use a prime mark: A′, B′, C′ (read "A prime").

Two Families of Transformations

Family	Preserves	Types	Result
Rigid (Isometry)			
Non-Rigid			

Big idea: If size stays the same, the figures are _____. If size changes but shape stays the same, they are _____.

2. Translations (Slides)

Translation: Every point slides the _____ distance in the _____ direction. The shape does NOT rotate or flip.

The Rule

$(x, y) \rightarrow (x + h, y + k)$

- **h** = horizontal shift. + moves _____, – moves _____
- **k** = vertical shift. + moves _____, – moves _____

Worked Example: Triangle ABC has vertices A(1, 2), B(4, 2), C(2, 5). Translate by (x – 3, y + 1).

Step 1 — Identify h and k: h = _____, k = _____

Step 2 — Apply to A(1, 2): (1 + ____, 2 + ____) = (____, ____) → A′ = (____, ____)

Step 3 — Apply to B(4, 2): (4 + ____, 2 + ____) = (____, ____) → B′ = (____, ____)

Step 4 — Apply to C(2, 5): (2 + ____, 5 + ____) = (____, ____) → C′ = (____, ____)

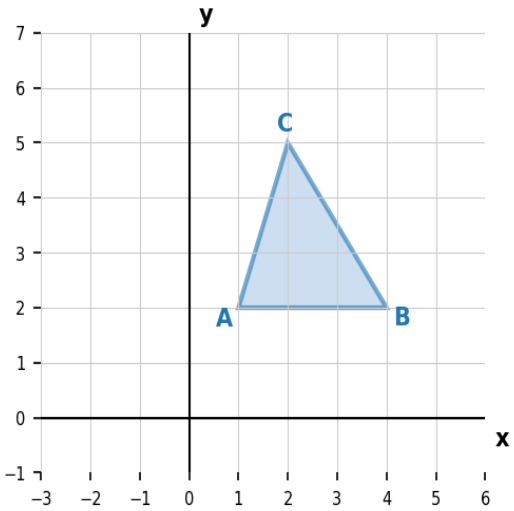


Image vertices: A′ _____ B′ _____ C′ _____

Plot the triangle and its image on the grid above.

3. Reflections (Flips)

Reflection: Every point flips across a _____ (the mirror line). Each point lands the _____ distance from the mirror, on the _____ side.

This year, you reflect over the x-axis and the y-axis.

The Rules

Line of Reflection	Rule	(3, 4) →	What Changes
x-axis			
y-axis			

Memory trick: The coordinate that *matches* the axis name stays the same. Reflect over the x-axis → x stays. Reflect over the y-axis → y stays.

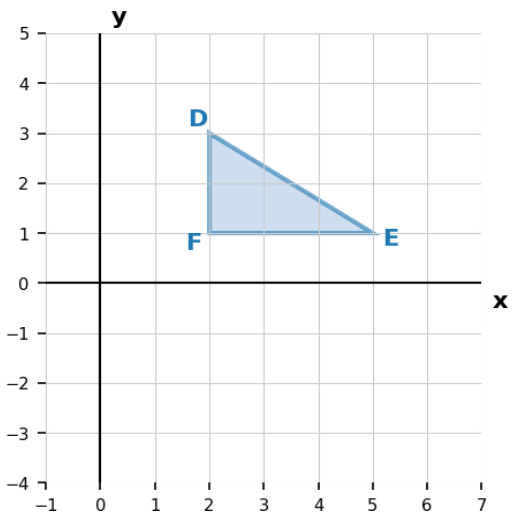
Worked Example: Reflect triangle DEF with D(2, 3), E(5, 1), F(2, 1) over the x-axis.

Step 1 — Rule: (x, y) → (_____ , _____). Keep x; flip y.

Step 2 — Apply to D(2, 3): D′ = (_____, _____)

Step 3 — Apply to E(5, 1): E′ = (_____, _____)

Step 4 — Apply to F(2, 1): F′ = (_____, _____)



Plot the triangle and its image on the grid above.

4. Rotations (Turns)

Rotation: Every point turns a fixed angle around a _____. In this unit, the center is always the _____.

This year, you rotate 90°, 180°, and 270° about the origin.

Direction matters! A rotation can be counterclockwise (CCW) or clockwise (CW). Always check which direction the problem asks for.

The Rules (About the Origin)

Rotation	CCW Rule	CW Rule	(3, 4) CCW →	(3, 4) CW →
90°				
180°				
270°				

Watch out: 180° is the only rotation where x and y are NOT swapped. For 90° and 270°, swap x and y FIRST, then apply the sign rule.

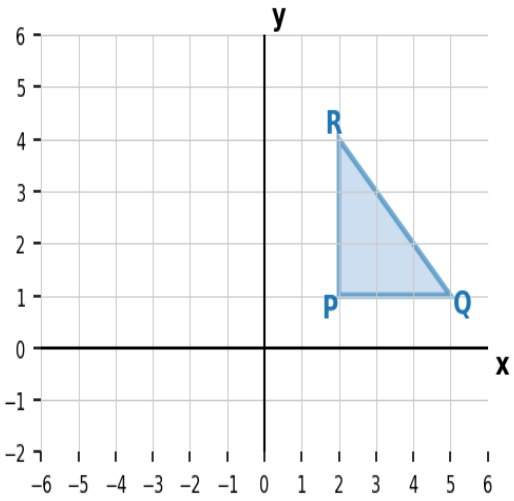
Worked Example: Rotate triangle PQR with P(2, 1), Q(5, 1), R(2, 4) 90° CCW about the origin.

Step 1 — Rule: 90° CCW: (x, y) → (____ , ____)

Step 2 — Apply to P(2, 1): P′ = (____, ____)

Step 3 — Apply to Q(5, 1): Q′ = (____, ____)

Step 4 — Apply to R(2, 4): R′ = (____, ____)



5. Dilations (Resize)

Dilation: A transformation that _____ a figure by a scale factor from a center point (usually the origin). The shape stays the same; the size changes.

The Rule (Centered at the Origin)

(x, y) → (_____ , _____)

Multiply both coordinates by the scale factor **k**.

What k Tells You

Scale Factor k	Effect	Image vs. Pre-image
k > 1		
k = 1		
0 < k < 1		

Watch out: Dilation is the only transformation in this unit that is NOT rigid. Pre-image and image are _____, but NOT _____ (unless k = 1).

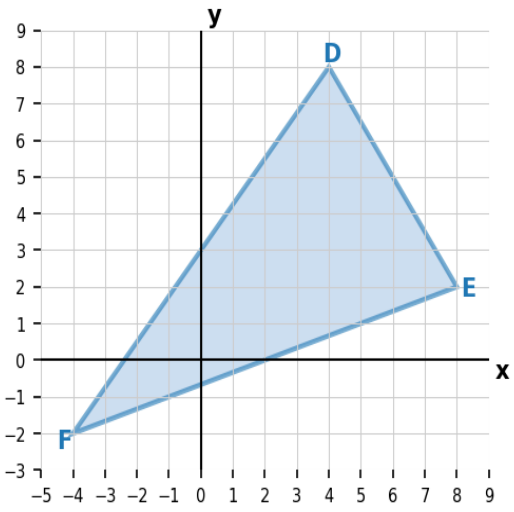
Worked Example: Triangle DEF has D(4, 8), E(8, 2), F(–4, –2). Dilate by k = 1/2 (centered at origin).

Step 1 — Rule: (x, y) → (_____ , _____)

Step 2 — Apply to D(4, 8): ((1/2)(4), (1/2)(8)) = (____, ____) → D′ = (____, ____)

Step 3 — Apply to E(8, 2): ((1/2)(8), (1/2)(2)) = (____, ____) → E′ = (____, ____)

Step 4 — Apply to F(–4, –2): ((1/2)(–4), (1/2)(–2)) = (____, ____) → F′ = (____, ____)



6. Congruence & Similarity

Congruent figures: Same size AND same shape. You can map one to the other using only _____ transformations (translations, reflections, rotations).

Similar figures: Same shape, possibly different size. You need rigid transformations AND a _____.

Decision rule:

- Used ONLY translations, reflections, rotations? → _____
- Used a dilation in the sequence? → _____ (unless k = 1)

What Gets Preserved?

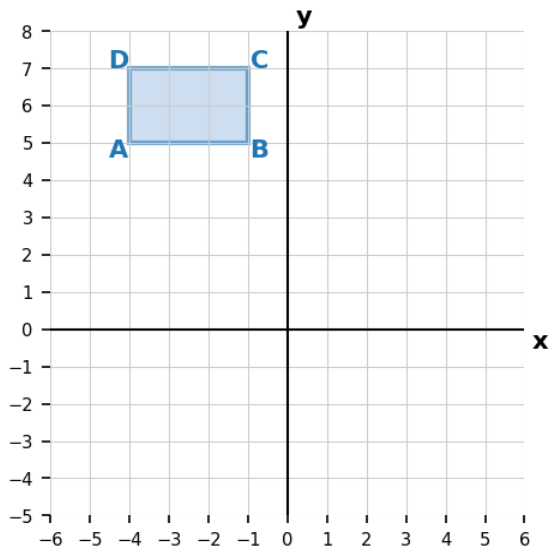
Property	Translations	Reflections	Rotations	Dilations
Side lengths				
Angle measures				
Shape				
Size				

Notice: Only dilations change _____. _____ stay the same in every transformation — that's why the shape never distorts.

✓ Practice Problems

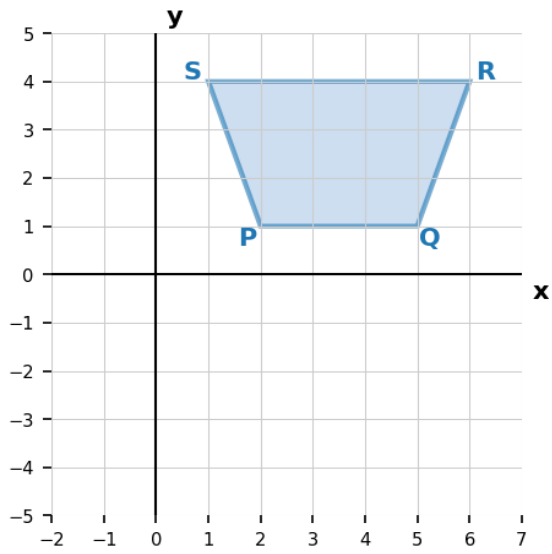
Apply what you learned. Plot if asked. Show your work on the writing lines.

Problem 1. Rectangle ABCD has vertices A(–4, 5), B(–1, 5), C(–1, 7), D(–4, 7). Translate the rectangle by the rule (x + 6, y – 8). Write the new coordinates A′, B′, C′, D′.



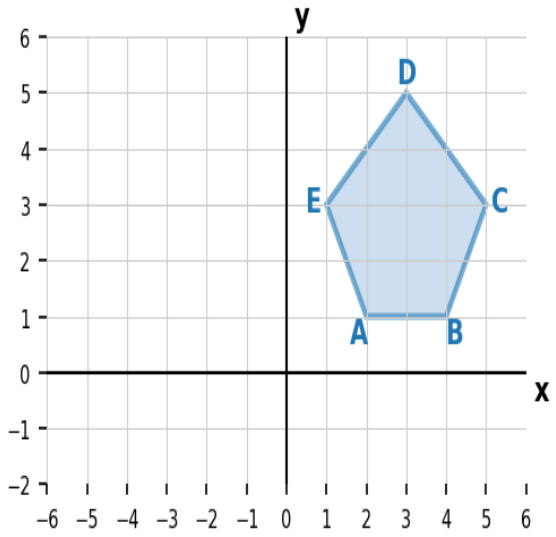
Answer / Work:

Problem 2. Quadrilateral PQRS has vertices P(2, 1), Q(5, 1), R(6, 4), S(1, 4). Reflect the quadrilateral over the x-axis. Write the new coordinates P′, Q′, R′, S′.



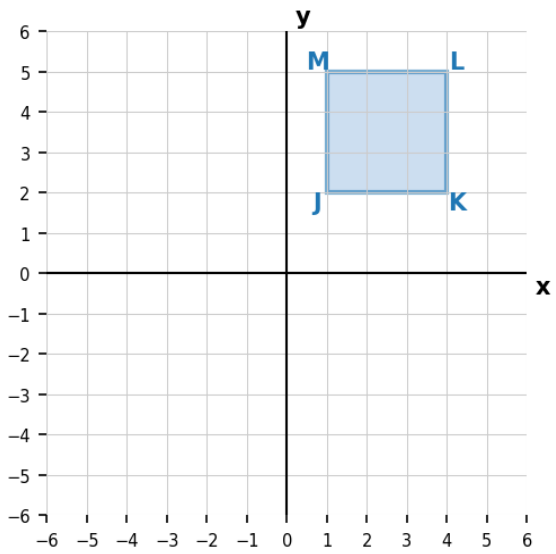
Answer / Work:

Problem 3. Pentagon ABCDE has vertices A(2, 1), B(4, 1), C(5, 3), D(3, 5), E(1, 3). Reflect the pentagon over the y-axis. Write the new coordinates A′, B′, C′, D′, E′.



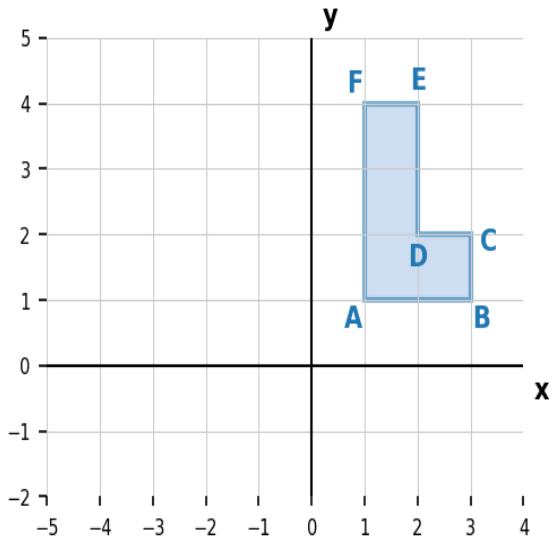
Answer / Work:

Problem 4. Rectangle JKLM has vertices J(1, 2), K(4, 2), L(4, 5), M(1, 5). Rotate the rectangle 180° about the origin. Write the new coordinates J′, K′, L′, M′.



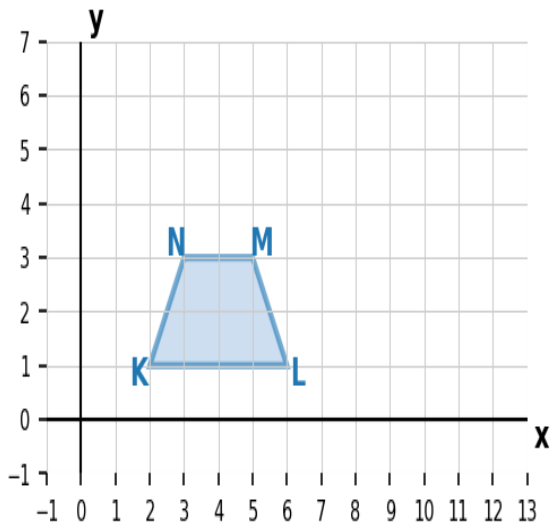
Answer / Work:

Problem 5. L-shape ABCDEF has vertices A(1, 1), B(3, 1), C(3, 2), D(2, 2), E(2, 4), F(1, 4). Rotate the L-shape 90° CCW about the origin. Write the new coordinates A′ through F′.



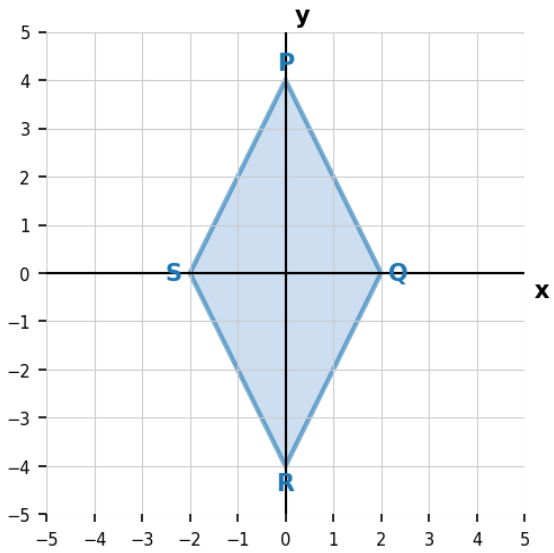
Answer / Work:

Problem 6. Trapezoid KLMN has vertices K(2, 1), L(6, 1), M(5, 3), N(3, 3). Dilate the trapezoid by scale factor k = 2 (centered at origin). Write the new coordinates K′, L′, M′, N′.



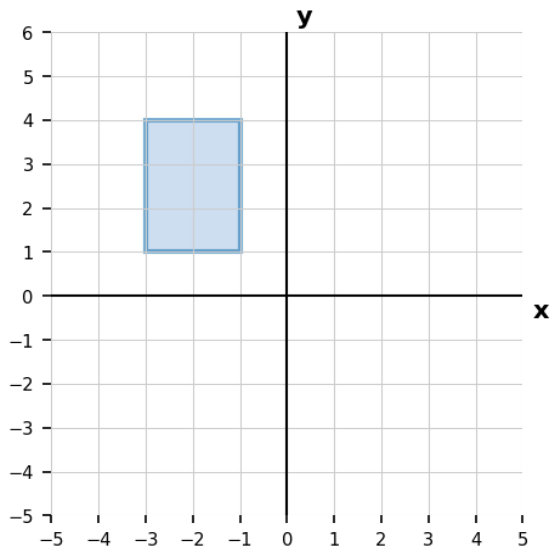
Answer / Work:

Problem 7. Kite PQRS has vertices P(0, 4), Q(2, 0), R(0, –4), S(–2, 0). Dilate the kite by scale factor k = 1/2 (centered at origin). Write the new coordinates P′, Q′, R′, S′.



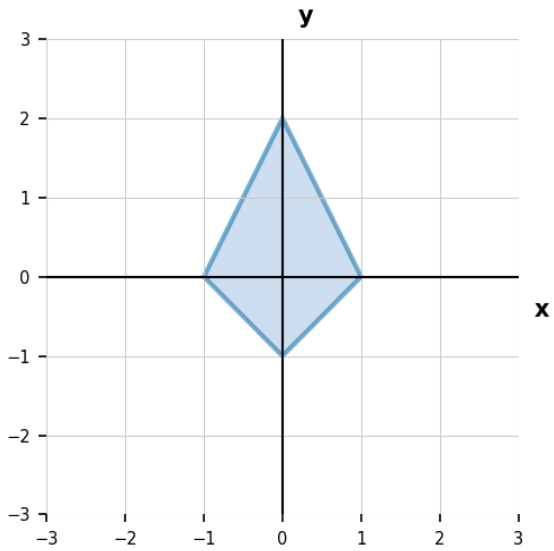
Answer / Work:

Problem 8. Rectangle X (shown) is reflected over the x-axis, then translated 3 units right. Are the pre-image and image congruent, similar, or neither? Explain.



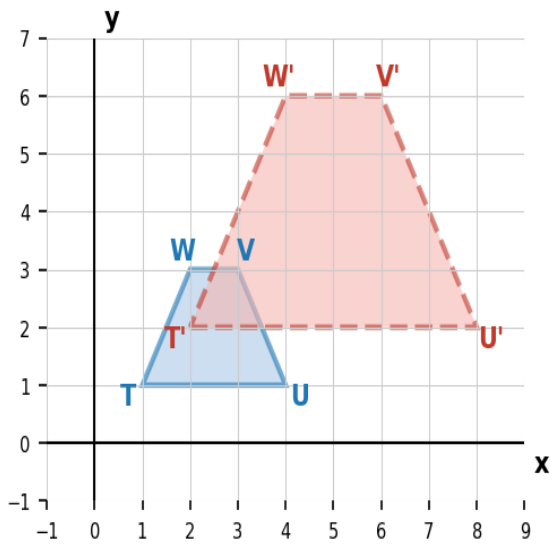
Answer / Work:

Problem 9. Kite Y (shown) is dilated by k = 3, then rotated 90° CCW. Are the pre-image and image congruent, similar, or neither? Explain.



Answer / Work:

Problem 10. Challenge: Trapezoid TUVW has T(1, 1), U(4, 1), V(3, 3), W(2, 3). After a transformation, the image is T′(2, 2), U′(8, 2), V′(6, 6), W′(4, 6). What transformation was applied? Explain how you know.



Answer / Work: